

Assessment 5

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1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Cloud **auto-scaling** automatically increases or decreases the number of servers based on traffic demand, while **load balancing** distributes incoming user requests across multiple servers.

How it Handles Traffic Spikes

- During sales, auto-scaling adds more instances
- Load balancer distributes traffic evenly
- Prevents any single server from overloading

Reducing Downtime & Improving UX

- Avoids server crashes → **zero/minimal downtime**
- Faster response time → **better user experience**
- Ensures consistent availability

Role of Monitoring Tools

- Track CPU, memory, and traffic
- Trigger auto-scaling policies
- Detect failures and reroute traffic

Real-World Example

During **Amazon/Flipkart Big Billion Days**, traffic spikes massively. Auto-scaling adds servers instantly, and load balancers distribute users, ensuring smooth checkout without crashes.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive

backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case. Durability & Redundancy

- Data is replicated across multiple locations
- Uses distributed storage systems
- Ensures **high durability (e.g., 99.999999999%)**

Security (Encryption & Access Control)

- **Encryption:** Data encrypted at rest and in transit
- **Access Control:** IAM policies restrict access
- Prevents unauthorized data exposure

Cost Advantages

- No need for physical hardware
- Pay-as-you-use model
- Reduced maintenance cost

Disaster Recovery Example

- Backup stored in cloud
- In case of system failure or ransomware attack
- Data is restored from cloud backups quickly

Example: A company restores its database from AWS S3 after server crash.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

How CDN Improves Performance

- Stores cached content near users
- Reduces distance between user and server

Role of Edge Servers

- Located globally
- Deliver content faster
- Reduce latency significantly

Cloud Support

- Global data centers and CDN networks
- Automatic traffic routing to nearest server

Real-World Example

- Netflix or online gaming platforms use CDNs
- Users get fast streaming with minimal buffering

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Regulatory Compliance

- Cloud providers support standards like GDPR, ISO, PCI-DSS
- Provide audit reports and compliance tools

Role of Logging & Monitoring

- Track all user activities
- Detect suspicious behavior
- Enable auditing

Encryption

- Protects sensitive financial data
- Ensures secure transactions

Data Sovereignty

- Data stored in specific geographic regions
- Meets legal requirements

Example

- Bank processes transactions securely
- Data encrypted and stored in region-specific data centers
- Logs maintained for audits

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Containers vs VMs

Feature	Containers	Virtual Machines
Performance	Faster	Slower
Resource Usage	Lightweight	Heavy
Startup Time	Seconds	Minutes

Role of Orchestration (Kubernetes)

- Automates deployment
- Handles scaling and load balancing
- Manages container failures

Portability

- Containers run consistently across environments
- Works on any cloud (multi-cloud support)

Real-World Example

- E-commerce app with microservices:
 - User service
 - Payment service
 - Product service
 Each runs in separate containers, managed by Kubernetes